

Multistage Planning Model for Active Distribution Systems and Electric Vehicle Charging Stations Considering Voltage-Dependent Load Behavior

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Abstract—This work proposes a novel mixed-integer linear programming model for the medium-term multistage planning of active distribution systems and electric vehicle charging stations (EVCSS). Investment alternatives include the installation of conductors, capacitor banks, voltage regulators, dispatchable and nondispatchable distributed generation, energy storage units, and EVCSSs. Hence, the model identifies the best size, location, and installation time for the candidate assets under the uncertainty associated with electricity demand, energy prices, renewable energy sources, and EVCSSs' load profiles. Unlike classical planning approaches, conventional load is modeled as voltage-dependent. Besides, EVCSSs are planned by zones to optimize the coverage of the service provided to users of electric vehicles and to reduce the discrepancy between the geographical requirements and the optimal locations for the installation of EVCSSs in the system. EVCSSs' load profiles are calculated using a travel simulation algorithm based on real travel patterns that consider fast, slow, and residential chargers. Moreover, as another salient feature, constraints for CO₂ emissions are incorporated into the model. The resulting model is formulated as a stochastic scenario-based program, which is driven by the minimization of the total expected cost. Tests are conducted using a 69-node system to demonstrate the effectiveness of the proposed model.

Index Terms—Active distribution systems, electric vehicle charging stations, mixed-integer linear programming, multistage planning, voltage-dependent load model.

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NOMENCLATURE

Sets and Indices

Ω^B, ij, ji	Set/indices of branches
Ω^C, a	Set/index of types of conductors
Ω^{ES}, e	Set/index of types of energy storage (ES) units
Ω^{EV}, v	Set/index of types of electric vehicles' (EVs) chargers
Ω^G, g	Set/index of types of dispatchable distributed generation (DG) units
Ω^{GS}, w	Set/index of groups of scenarios
Ω^N, i	Set/index of nodes
Ω^{RS}, r	Set/index of types of nondispatchable renewable energy sources (RESs) units
Ω^S, s	Set/index of scenarios
Ω^{SS}, i	Set/index of substation nodes
Ω^T, t	Set/index of stages
Ω^Z, z	Set/index of zones
c	Index of capacitor banks' (CBs) modules
h	Index used to obtain the type of conductor or voltage regulator (VR) operating at a stage according to the corresponding investment variable
l	Index of blocks of the piecewise linearization.

Parameters

B^{CB}	Susceptance of a CB's module
$c_{a,a}^C$	Cost of replacing conductor of type a by type a
c_e^{CB}, j^{CB}	Investment/installation costs of CBs
$c_e^{ES}, j_e^{ES}, o_e^{ES}$	Investment/installation/operation costs of ES units
c_v^{EV}, j^{EV}	Investment cost of EVs' chargers/installation cost of electric vehicle charging stations (EVCSSs)
c_g^G, o_g^G	Investment/operation costs of dispatchable DG units
$c_r^{RS}, j_r^{RS}, o_r^{RS}$	Investment/installation/operation costs of nondispatchable RES units
$c_a^{VR}, \mathcal{R}^{VR}$	Cost of VRs/regulation of VRs

$\varepsilon_i^{SS}, \varepsilon_g^G, \varepsilon_r^{RS}$	CO ₂ emissions rates of substations/dispatchable DG units/nondispatchable RESs units
$\mathcal{F}_{r,s}^{RS}$	Generation factor of nondispatchable RESs units in a scenario.
$\mathcal{G}_s$	Group to which a scenario belongs
$\bar{I}_a$	Current capacity of conductors
IC_t, OC_t	Investment/operation costs
L_{ij}	Length of branches
$\mathcal{M}_{ij,l}^{\hat{s}}$	Slope of a block of the piecewise linearization
$\bar{N}_i^{CB}$	Limit for the number of CBs' modules at nodes
$\bar{N}_t^{CB}$	Limit for the number of CBs in the system
$\bar{N}_{i,e}^{ES}$	Limit for the number of ES units at nodes
$\bar{N}_{e,t}^{ES}$	Limit for the number of ES units in the system
$\bar{N}_i^{EV}$	Limit for the number of EVs' chargers at nodes
$\bar{N}_t^G$	Limit for the number of dispatchable DG units in the system
$\bar{N}_{i,r}^{RS}$	Limit for the number of nondispatchable RES units at nodes
$\bar{N}_{r,t}^{RS}$	Limit for the number of nondispatchable RES units in the system
$\bar{N}_t^{VR}$	Limit for the number of VRs in the system
$\mathcal{O}_t^E$	Price of energy
$P_{i,t,s}^D, Q_{i,t,s}^D$	Active/reactive power demands at nominal voltage
$\bar{P}_e^{ES}$	Nominal capacity of ES units
$\mathcal{P}_{z,v,t,s}^{EV}$	Total active power demand by EVs' chargers per zone
$\bar{P}_v^{EV}$	Nominal capacity of EVs' chargers
$\underline{PF}_g^G, \overline{PF}_g^G$	Capacitive/inductive power factor limits of dispatchable DG units
$\underline{PF}_r^{RS}, \overline{PF}_r^{RS}$	Capacitive/inductive power factor limits of nondispatchable RES units
R_a, X_a, Z_a	Resistance/reactance/impedance of conductors
$\bar{S}_g^G$	Nominal capacity of dispatchable DG units
$\bar{S}_r^{RS}$	Nominal capacity of nondispatchable RES units
$\bar{S}_i^{SS}$	Nominal capacity of substations
$\mathcal{U}_t$	Minimum penetration of RESs
V	Nominal voltage of the system
$\bar{V}, \underline{V}$	Upper/lower voltage limits
$\mathcal{V}_{i,t,s}$	Estimate of the voltage magnitude
$\bar{\mathcal{X}}_{ij,a}^C$	Binary parameter indicating the initial type of conductor on a branch
$\bar{\mathcal{X}}_{i,g}^G$	Binary parameter that indicates if a node is a candidate for the installation of a dispatchable DG unit
$\mathcal{Z}_i$	Zone in which a node belongs
$\Gamma_i^Z, \Gamma_i^I, \Gamma_i^P$	Active load components of constant impedance/current/power
Δ_s^T	Duration of a scenario

Θ	Maximum allowable amount of CO ₂ emissions
Λ	Number of discretization blocks used in the piecewise linearization
Ξ_e^{ES-}, Ξ_e^{ES+}	ES units' charge/discharge efficiencies
Π	Annual interest rate
$\Upsilon_t^I, \Upsilon_t^F$	Initial/final years of each planning stage
$\bar{\Phi}_{ij}^{\hat{s}}$	Length of a block of the piecewise linearization
$\Psi_i^Z, \Psi_i^I, \Psi_i^P$	Reactive load components of constant impedance/current/power.

Continuous Variables

$i_{ij,a,t,s}^{SQ}, \hat{i}_{ij,t,s}^{SQ}$	Squared current value on conductors/branches
$p_{ij,a,t,s}, q_{ij,a,t,s}$	Active/reactive power flows on conductors
$\hat{p}_{ij,t,s}, \hat{q}_{ij,t,s}$	Total active/reactive power flows on branches
$\hat{p}_{ij,t,s}^+, \hat{p}_{ij,t,s}^-$	Nonnegative variables used in the linearization of the total active power flows on branches
$p_{i,e,t,s}^{ES-}, p_{i,e,t,s}^{ES+}$	Nonnegative variables that represent the active power stored/injected by ES units
$p_{i,v,t,s}^{EV}$	Active power demanded by EVCSs
$p_{i,g,t,s}^G, q_{i,g,t,s}^G$	Active/reactive power from dispatchable DG units
$p_{i,r,t,s}^{RS}, q_{i,r,t,s}^{RS}$	Active/reactive power from nondispatchable RES units
$p_{i,t,s}^{SS}, q_{i,t,s}^{SS}$	Active/reactive power from substations
$\hat{q}_{ij,t,s}^+, \hat{q}_{ij,t,s}^-$	Nonnegative variables used in the linearization of the total reactive power flows on branches
$q_{i,c,t,s}^{CB}, \hat{q}_{ij,t,s}^{CB}$	Reactive power injected by CBs' modules/CBs
$v_{i,t,s}, v_{i,t,s}^{SQ}$	Nodal voltage magnitude/its squared value
$\xi_{ij,t,s}^{VR}$	Difference of the squared values of the voltage magnitudes at the terminal nodes of a VR
$\phi_{ij,l,t,s}^{\hat{p}}, \phi_{ij,l,t,s}^{\hat{q}}$	Discretization variables for the total active/reactive power flows of the piecewise linearization.

Integer-Valued Continuous Variables

$n_{i,t}^{CB}$	Number of CB's modules present at node i , stage t
$\psi_{i,t}^{CB}$	Number of CB's modules installed at node i , stage t
$\psi_{i,e,t}^{ES}$	Number of ES units of type e installed at node i , stage t
$\psi_{i,v,t}^{EV}$	Number of EVs' chargers of type v installed at node i , stage t
$\psi_{i,r,t}^{RS}$	Number of nondispatchable RES units of type r installed at node i , stage t .

Binary-Valued Continuous Variables

$x_{ij,a,t}^C$	Indicates the presence of a conductor of type a on branch ij , stage t
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$\hat{x}_{ij,a,t}^C$	Indicates the type a of new conductor operating on branch ij , stage t
$x_{ij,t}^C$	Indicates in which branches ij , stage t , new conductors are operating
$x_{ij,a,t}^{VR}$	Indicates the presence of a VR of type a on branch ij , stage t
$x_{i,t}^{CB}$	Indicates the installation of a CB at node i , stage t
$x_{i,e,t}^{ES}$	Indicates the installation of ES units of type e at node i , stage t
$x_{i,t}^{EV}$	Indicates the installation of an EVCS at node i , stage t
$x_{i,g,t}^G$	Indicates the installation of dispatchable DG unit of type g at node i , stage t
$x_{i,r,t}^{RS}$	Indicates the installation of nondispatchable RES units of type r at node i , stage t .

Binary Variables

$x_{i,c,t}^{CB}$	Indicates the presence of CB's module of type c at node i , stage t
$x_{i,g,t}^G$	Indicates the presence of dispatchable DG unit of type g at node i , stage t
$y_{i,t}^{CB}$	Indicates the presence of a CB at node i , stage t
$y_{i,t}^{ES}$	Indicates the presence of an ES unit at node i , stage t
$y_{i,t}^{EV}$	Indicates the presence of an EVCS at node i , stage t
$y_{i,t}^{RS}$	Indicates the presence of a nondispatchable RES unit at node i , stage t
$x_{ij,a,t}^C$	Decision on installing a conductor of type a on branch ij , stage t
$x_{ij,a,t}^{VR}$	Decision on installing a VR of type a on branch ij , stage t .

Integer Variables

$n_{i,e,t}^{ES}$	Number of ES units of type e present at node i , stage t
$n_{i,v,t}^{EV}$	Number of EVs' chargers of type v present at node i , stage t
$n_{i,r,t}^{RS}$	Number of nondispatchable RES units of type r present at node i , stage t .

I. INTRODUCTION

CONCERNS on global warming and the depletion of fossil fuels are causing the electricity and transportation industries, the largest consumers of fossil fuels, to change their modus operandi in order to reduce environmental pollution and dependence on oil [1]. In this way, the adoption of electric vehicles (EVs) has emerged as a potential alternative to reduce CO₂ emissions and decrease the consumption of liquid fuels in large metropolises. However, the large-scale deployment of EVs calls for the installation of electric vehicle charging stations (EVCSs) that may lead to a strong increase in electricity demand [2]. Thus, the distribution networks must be properly planned to be able to support the new electricity demands. Moreover, the notable environmental benefits of distributed generation (DG)

based on renewable energy has driven its installation in distribution systems around the world. Unfortunately, the variability and uncertainty associated with the primary sources (wind speed, solar irradiation) of renewable DGs hinder the distribution system planning (DSP) [3]. The effect of uncertainty may be partially compensated for by the installation of energy storage (ES) units, which increases the manageability of the grid [4]. On the other hand, special attention must be paid to the installation of renewable DGs in the network, as inadequate planning for such devices may result in several issues to the network operation [5]. Although the installation of large-scale renewable generation is done at the transmission level, the integration of distributed generation at the distribution level is also possible and provides several benefits to the network operation, such as voltage profile improvement, technical losses reduction, reliability improvement, etc. Also, it postpones the necessity of investments in the reinforcement of the distribution and transmission network, since the load is supplied locally [6].

Relevant examples of research addressing the installation of EVCSs and renewable DGs within the DSP using a multistage approach are presented in [7]–[10]. It is worth mentioning that a multistage planning approach allows for greater investment control by enabling the planner to identify the installation time of candidate assets so that the load increase is optimally satisfied [11]. A complete review of [7]–[10] is presented below.

In [7], a multi-objective stochastic programming model is used to include the size and location of fast and slow chargers and battery exchange stations within the DSP problem. The objective function minimizes the investment and operating costs of the network while maximizing the annual traffic flow captured by the EVCSs. The proposed model is solved by a multi-objective evolutionary algorithm based on a decomposition strategy.

In [8], a robust programming model is proposed to include the installation of fast, slow, and residential chargers, as well as the deployment of renewable DG units in the DSP. Investments in capacitor banks (CBs) are also incorporated. Probabilistic constraints are used to address the uncertainty of the problem. The objective function minimizes the investment, operating, and energy loss costs throughout the planning horizon. The resulting mixed-integer linear programming model (MILP) is solved using an optimization solver. An MILP model, similar to that presented in [8], is shown in [9]. The difference is that in [9], the EVCSs' load profiles are estimated using EVs' driving patterns, uncertainty is addressed using a scenario-based stochastic programming model, and investments in ES units are considered.

Finally, in [10], a scenario-based stochastic programming model is proposed to solve the DSP problem where investments in renewable DG units, ES units, and EVCSs are jointly considered. The EVCSs' load profiles are calculated using a vehicular transportation algorithm.

The works described in [7]–[10] model the load behavior using a constant power representation, as it is more practical and easier to implement. However, several studies on the operation of medium-voltage distribution systems have shown

TABLE I
SUMMARY OF LITERATURE REVIEW ON MULTISTAGE PLANNING FEATURES

Reference	Conductors replacement	Investment Alternatives										CO ₂ emissions constraints	EVs adoption by zones	Voltage dependent load model	Solution technique
		CBs		VRs		DG		ES units		EVCS					
		Size	Location	Size	Location	Size	Location	Size	Location	Size	Location				
[7]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✗	Metaheuristic
[8]	✓	✓	✓	✗	✗	✗	✓	✗	✗	✓	✓	✗	✗	✗	MILP
[9]	✓	✓	✓	✗	✗	✗	✓	✓	✓	✓	✓	✗	✗	✗	MILP
[10]	✓	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✗	✗	✗	MILP
This work	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	MILP

that the behavior of load depends on voltage [12] and that correctly modeling this behavior allows for significant reductions in energy consumption, which may have an important impact on the investment plans [13], [14]. Within this context, the incorporation of investment decisions in voltage control equipment, such as CBs and voltage regulators (VRs), in the DSP model becomes more important.

Moreover, the availability of infrastructure for charging EVs' batteries is one of the factors that have a significant impact on the adoption of EVs [15]. Therefore, when planning the location of EVCSs, it should provide EVs with the greatest possible coverage. In [7]–[10], the EVCSs are planned considering only their optimal location in the energy system. However, this location does not necessarily correspond with the required geographical necessities since the adoption of EVs is heterogeneous and varies in each zone of a city according to the socio-economic characteristics of the population [16]. Although methods for planning EVCSs with an explicit representation of the vehicular traffic network have been proposed, e.g., [17]–[19], these methods have neither considered multistage planning nor all the investment alternatives considered in this work. Thus, to reach a compromise between the optimal location in the system and the suitable geographical location for the installation of EVCSs, this work proposes to divide the coverage area of the distribution network into zones and to plan the sizing and location of the EVCSs according to the EVs' charging requirements in each zone. In the proposed approach, the vehicular traffic network is used to estimate the travel patterns, in the preprocessing stage of the input data of the model, therefore, being implicitly considered in the proposed formulation.

Furthermore, the models presented in [7]–[10] do not use an explicit formulation to limit the emissions of CO₂ from the energy system. Although the aim of deploying EVs is to reduce the environmental impact, this purpose could be hindered by a high increase in the emissions of CO₂ from the network due to an increase in the demand for electricity needed to charge the batteries, and as a result, distribution companies could face serious penalties for violating environmental regulations that have been implemented worldwide to deal with global warming [20].

This paper proposes a new multistage planning model for active distribution systems and EVCS considering voltage-dependent load behavior. Here, the model presented in [21] is extended in a non-trivial way to adopt a multistage planning framework while also incorporating investment decisions

on EVCS. As a result, the case studies and contributions substantially differ from [21]. Moreover, unlike [7]–[10], the investment decisions considered in this work include the sizing, location, and installation time of conductors, CBs, VRs, dispatchable DG units, renewable DG units, ES units, and EVCSs, all at the same time. The EVCSs' load profiles for each zone are obtained using a travel simulation algorithm based on real travel patterns. Moreover, a set of constraints related to CO₂ emissions is used to prevent that EVs' charging becomes a grid emissions problem. The system operation is represented using a linearized ac power flow formulation. Uncertainties, such as electricity demand, nondispatchable renewable generation, energy prices at the substations, and EVCSs' load profiles are addressed through a scenario-based stochastic programming model. The DSP problem is a mixed-integer nonlinear programming (MINLP) problem. Therefore, linearization techniques are used to convert the original problem into an approximate MILP model, which can be written in mathematical modeling languages, such as AMPL, and solved using commercial solvers, such as CPLEX.

Table I summarizes the main differences between the proposal presented in this paper and the research aimed at addressing the joint planning of active distribution networks, EVCSs, and renewable DGs using a multistage approach, which was presented in [7]–[10]. Thus, symbols "✓" and "×" respectively indicate whether a particular aspect is considered or not. Therefore, the main contributions of this work are:

- A new scenario-based MILP model is proposed for the multistage planning of active distribution systems and EVCSs considering multiple planning options, voltage-dependent load behavior, and CO₂ emissions constraints. The advantage of considering multiple investment alternatives is that the planning model has more options for selecting the combination of network assets that provides the greatest benefits. In addition, modeling the load as voltage-dependent enables the planning model to use the conservation voltage reduction strategy that consists of reducing network voltage levels to reduce the load and thus achieve reductions in energy consumption [12]. Finally, using CO₂ emissions restrictions in the DSP prevents the distribution company from being penalized for exceeding the permitted emissions limits;
- A new approach for the planning of EVCSs is presented. To that end, the coverage area of the distribution network is divided into zones. EVs' charging requirements for each zone are estimated using a novel algorithm based

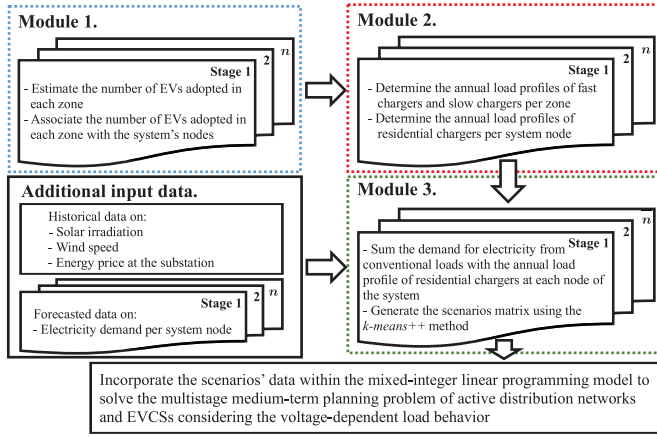


Fig. 1. Framework of the method for generating the scenarios.

on real travel patterns, which is specifically tailored to accommodate the division in zones. In this way, the population census data can be used to estimate the number of EVs adopted in each zone. This allows the proposed optimization problem to determine the best size, location, and installation time of EVCSs for each zone. The proposed approach aims to reduce the existing discrepancy between the optimal location in the distribution system and the appropriate geographic location for the EVCSs installation.

The aspects described above have not been explored as alternatives to solving the problem addressed in this work, and the results show that considering them has a significant impact on the total cost of the DSP.

The remainder of this paper is organized as follows: Section II shows the method used to estimate the EVs' charging demand and the uncertain parameters through scenarios. Section III illustrates the mathematical formulation of the DSP problem. Section IV presents the case study and results for a 69-node system. Finally, Section V concludes this work.

II. EVs' CHARGING DEMAND AND UNCERTAINTY MODELING

The input data for the proposed optimization model are calculated using three modules, as explained below. Fig. 1 shows a diagram of the proposed framework to model the behavior of the uncertain parameters.

It should be noted that modeling the traffic network within the proposed model would render the resulting problem essentially intractable through optimization, thereby requiring the use of heuristics or repeated simulations. This modeling limitation notwithstanding, the solution of the multistage planning model for active distribution systems and electric vehicle charging stations is acceptable for distribution planning purposes [7]–[10] and provides the planners and regulators with a first estimate of a cost-effective and environmentally-friendly expansion scheme.

A. Module 1: Estimation of EVs' Adoption Per Zone

First, the module calculates the total number of EVs adopted at each stage of the planning horizon on the basis of a

percentage of the total number of existing vehicles in the entire coverage area of the distribution network.

Then, the module makes a selection of zones for the spatial distribution of the results of the EVs adopted at each stage. In each iteration, only one EV is associated with the selected zone. The probability of a zone being selected depends on the number of residential consumers in the zone, the average ownership of vehicles per household in the zone, and the socioeconomic characteristics of the population in the zone, which can be extracted from the population census data.

Finally, the module makes a random selection of the network nodes to link the EVs adopted in each zone with the power system. This association is necessary to identify the nodes to which the load profiles of each residential charger should be added. In each iteration, only one EV is associated with the selected node. The probability of a node being selected depends on the number of residential consumers connected to it.

For large-sized real distribution systems, when census data is available, an alternative to be used in this module is the approach presented in [16], which considers the city map division by census tracts to determine the spatial-temporal adoption of EVs. Note that, in this approach, the zones of the city are readily available as census tracts. Moreover, according to the planner's criterion, the census tracts can be merged for covering larger areas.

B. Module 2: Estimation of EVs' Charging Demand Per Zone

With the results provided by Module 1, i.e., the number of EVs adopted in each zone and associated with each node, Module 2 estimates the EVs' charging demand per zone.

The EVs' charging demand is calculated using a decision algorithm that is based on real travel patterns and assumptions about when and where EVs should be charged. The algorithm was implemented in MATLAB and it is based on realistic vehicle statistics extracted from [22], which includes data for each travel distance, start and end times, month, day of the week, the purpose of the trip, and location of the trip. Note that, the vehicular traffic network is considered in this procedure. This work assumes that the EVs' travel patterns are the same as for traditional fossil-fueled vehicles. Fig. 2 shows the flowchart of the algorithm proposed to calculate the load profiles of EVs by zones. The main steps of the algorithm are the following:

- i. Assign a battery capacity (24, 35, or 50 kWh) and an initial state of charge (SOC) to each EV ν (between 50–100%).
- ii. Assign a daily travel profile to each EV ν , month m , day d in which the start time of the first trip of the day, the duration of each trip, distance traveled on each trip, and the waiting time for the next trip are specified. Consider that there are days when the EV ν does not travel.
- iii. Check the SOC of the battery continuously during each trip $\mathcal{T}$. If the SOC is greater than a percentage $\mathfrak{i}$ of the capacity, the EV will continue the trip and the SOC will be updated and computed. Otherwise, EV ν has to be charged.

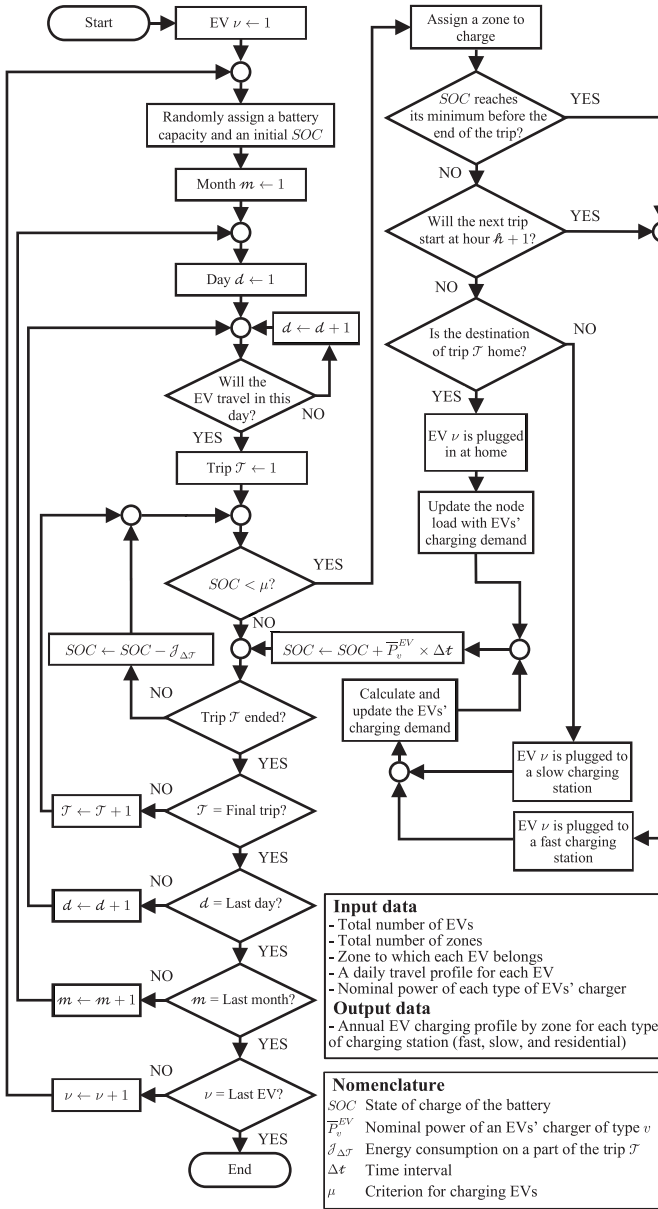


Fig. 2. Flowchart of the charging decision model for EVs.

iv. If EV ν has to be charged, determine the zone where it will be charged. EV ν can be charged in any zone, but the priority is to charge it in the zone to which it belongs. The zone will not be completely randomly assigned since the zone to which the EV belongs has the highest probability of being assigned while neighboring zones and other zones have lower probabilities. Determine whether EV ν should be charged with a fast charger, a slow charger, or at home. Calculate and update the charging profile of the EVs by zone, as well as the SOC of the battery. If the EV is plugged-in at a slow charger or at home, it must be charged until the battery has reached 90% of its capacity or until the time for the next trip. If EV ν is to be plugged-in at a fast charger, the algorithm randomly selects a charging duration between 10–60 minutes. The charging ends when

this time is over or when the battery reaches 90% of its capacity.

v. Repeat these steps for all days, months, and EVs.

This work considers a 5-kW power capacity for slow chargers and residential chargers and a 10-kW power capacity for fast chargers. The algorithm calculates the total EVs' charging demand for all zones per hour, day, month, and year.

C. Module 3: Characterization of the Uncertainties

This work considers five uncertain parameters: electricity demand, wind speed, solar irradiation, energy prices, and EVCSs' load profiles in each zone. The behavior of the uncertain parameters is characterized by using a stochastic scenario-based programming framework. The scenarios are based on historical data and are classified into four groups based on the seasons of the year (spring, summer, fall, and winter). Each group is then divided into two sub-groups (day/night). To divide the data from each sub-group into k clusters, the k -means++ algorithm [23] is used. Thus, a set of $(4 \times 2 \times k)$ scenarios represents the annual behavior of the uncertain parameters. It should be noted that the historical data of wind speed, solar irradiation, and energy prices remain constant at all stages of the planning horizon, while the electricity demand and EVCS load profiles in each zone vary from one stage to another stage, depending on the load growth forecast and the EV adoption forecast at each stage.

III. MATHEMATICAL MODELING OF THE PROBLEM

The proposed model is based on the following premises: (i) the planning horizon is divided into $|\Omega^T|$ stages, (ii) representative scenarios are used to model the annual variation of the uncertain parameters under consideration; (iii) a linearized ac model is used to represent the steady-state operation of the network; (iv) the distribution network operates with a radial topology; (v) a ZIP polynomial representation is used to model the conventional load as voltage-dependent; (vi) there is an environmental policy that penalizes excess of CO₂ emissions from the distribution system; and (vii) a centralized planning strategy is being considered. Regardless of who owns the distributed energy resources (DERs) and EVCSs, the proposed model provides valuable information on the best investment plan in terms of both economic and environmental factors. Such data can be used to develop appropriate incentive policies for cases in which DERs and EVCSs are owned by independent agents. However, such incentive policies are beyond the scope of this work.

A. Objective Function

The objective function (1) minimizes the present value of the total expected costs (investment cost and operating cost). The annual investment cost, IC_t , shown in (2), consists of seven terms, namely, the investments in (i) replacement of conductors, (ii) VRs, (iii) CBs, (iv) dispatchable DG units, (v) nondispatchable renewable energy sources (RESs), such as photovoltaic (PV) generation units and wind turbine (WT) generation units, (vi) ES units, e.g., battery banks, and (vii) the infrastructure for EVCSs, and fast chargers and slow chargers

at EVCSs. Moreover, the installation costs, $\mathcal{J}^{CB}$, $\mathcal{J}_r^{RS}$, $\mathcal{J}_e^{ES}$, and $\mathcal{J}^{EV}$ are considered to highlight that installing a quantity of a given network asset at separate nodes is more expensive than installing the same quantity of that network asset at the same node. In contrast, the operating cost, OC_t , shown in (3), comprises (i) the cost of the energy supplied by the substation, (ii) the generation cost of dispatchable DG units, (iii) the generation cost of nondispatchable RESs, and (iv) the operating cost of ES units.

$$\text{minimize } \sum_{t \in \Omega^T} (IC_t + OC_t) \quad (1)$$

$$IC_t = \frac{\Upsilon_t^F |\Omega^T| + 1 - t}{(1 + \Pi) \Upsilon_t^I} \times \left\{ \sum_{ij \in \Omega^B} \sum_{a \in \Omega^C} \left(\mathcal{C}_{a,a}^C L_{ij} \chi_{ij,a,t}^C + \mathcal{C}_a^{VR} \chi_{ij,a,t}^{VR} \right) + \sum_{i \in \Omega^N} \left[\left(\mathcal{J}^{CB} \chi_{i,t}^{CB} + \mathcal{C}_{i,t}^{CB} \psi_{i,t}^{CB} \right) + \sum_{g \in \Omega^G} \mathcal{C}_g^G \chi_{i,g,t}^G + \sum_{r \in \Omega^{RS}} \left(\mathcal{J}_r^{RS} \chi_{i,r,t}^{RS} + \mathcal{C}_r^{RS} \psi_{i,r,t}^{RS} \right) + \sum_{e \in \Omega^{ES}} \left(\mathcal{J}_e^{ES} \chi_{i,e,t}^{ES} + \mathcal{C}_e^{ES} \psi_{i,e,t}^{ES} \right) + \left(\mathcal{J}^{EV} \chi_{i,t}^{EV} + \sum_{v \in \Omega^{EV}} \mathcal{C}_v^{EV} \psi_{i,v,t}^{EV} \right) \right] \right\} \quad (2)$$

$$OC_t = \frac{1}{(1 + \Pi) \Upsilon_t^I} \left[\frac{(1 + \Pi)(\Upsilon_t^F - \Upsilon_t^I) - 1}{\Pi(1 + \Pi)(\Upsilon_t^F - \Upsilon_t^I)} \right] \times \sum_{i \in \Omega^N} \sum_{s \in \Omega^S} \Delta_s^T \left(\mathcal{O}_s^E p_{i,t,s}^{SS} + \sum_{g \in \Omega^G} \mathcal{O}_g^G p_{i,g,t,s}^G + \sum_{r \in \Omega^{RS}} \mathcal{O}_r^{RS} p_{i,r,t,s}^{RS} + \sum_{e \in \Omega^{ES}} \mathcal{O}_e^{ES} p_{i,e,t,s}^{ES+} \right) \quad (3)$$

The investment costs are brought to the present value considering an interest rate Π and that each stage t begins at year Υ_t^I . For the operation costs, it is considered a load increase in each stage, which comprises the period from Υ_t^I to Υ_t^F , and the costs are brought to the present value considering the interest rate Π .

B. Steady-State Operational System Constraints

The constraints related to the steady-state operation of the system are shown in (4)–(22).

$$\sum_{ji \in \Omega^B} \sum_{a \in \Omega^C} p_{ji,a,t,s} - \sum_{ij \in \Omega^B} \sum_{a \in \Omega^C} \left(p_{ij,a,t,s} + R_a L_{ij}^I \chi_{ij,a,t,s}^{SQ} \right) + p_{i,t,s}^{SS} + \sum_{g \in \Omega^G} p_{i,g,t,s}^G + \sum_{r \in \Omega^{RS}} p_{i,r,t,s}^{RS} + \sum_{e \in \Omega^{ES}} \left(p_{i,e,t,s}^{ES+} - p_{i,e,t,s}^{ES-} \right)$$

$$= \left(\Gamma_i^Z \frac{v_{i,t,s}^{SQ}}{V^2} + \Gamma_i^I \frac{v_{i,t,s}}{V} + \Gamma_i^P \right) p_{i,t,s}^D + \sum_{v \in \Omega^{EV}} p_{i,v,t,s}^{EV} \quad (4)$$

$$\sum_{ji \in \Omega^B} \sum_{a \in \Omega^C} q_{ji,a,t,s} - \sum_{ij \in \Omega^B} \sum_{a \in \Omega^C} \left(q_{ij,a,t,s} + X_a L_{ij}^I \chi_{ij,a,t,s}^{SQ} \right) + q_{i,t,s}^{SS} + \sum_{g \in \Omega^G} q_{i,g,t,s}^G + \sum_{r \in \Omega^{RS}} q_{i,r,t,s}^{RS} + \hat{q}_{i,t,s}^{CB} = \left(\Psi_i^Z \frac{v_{i,t,s}^{SQ}}{V^2} + \Psi_i^I \frac{v_{i,t,s}}{V} + \Psi_i^P \right) q_{i,t,s}^D \quad (5)$$

$$v_{i,t,s} = \sqrt{\frac{\bar{V} + V}{2}} + \frac{1}{2\sqrt{\frac{\bar{V} + V}{2}}} \left(v_{i,t,s}^{SQ} - \frac{\bar{V} + V}{2} \right) \quad \forall i \in \Omega^N, t \in \Omega^T, s \in \Omega^S \quad (6)$$

$$v_{i,t,s}^{SQ} - v_{j,t,s}^{SQ} + \xi_{ij,t,s}^{VR} = \sum_{a \in \Omega^C} \left[2L_{ij} (R_a p_{ij,a,t,s} + X_a q_{ij,a,t,s}) + Z_a^2 L_{ij}^2 \chi_{ij,a,t,s}^{SQ} \right] \quad (7)$$

$$v_{j,t,s}^2 \hat{\chi}_{ij,t,s}^{SQ} = \sum_{l=1}^{\Lambda} \mathcal{M}_{ij,l}^{\hat{S}} \left(\hat{\phi}_{ij,l,t,s}^{\hat{P}} + \hat{\phi}_{ij,l,t,s}^{\hat{Q}} \right) \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (8)$$

$$\hat{p}_{ij,t,s} = \hat{p}_{ij,t,s}^+ - \hat{p}_{ij,t,s}^- \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (9)$$

$$\hat{p}_{ij,t,s}^+ + \hat{p}_{ij,t,s}^- = \sum_{l=1}^{\Lambda} \hat{\phi}_{ij,l,t,s}^{\hat{P}} \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (10)$$

$$0 \leq \hat{\phi}_{ij,l,t,s}^{\hat{P}} \leq \bar{\Phi}_{ij}^{\hat{S}} \quad \forall ij \in \Omega^B, l \in \{1, \dots, \Lambda\}, t \in \Omega^T, s \in \Omega^S \quad (11)$$

$$\hat{q}_{ij,t,s} = \hat{q}_{ij,t,s}^+ - \hat{q}_{ij,t,s}^- \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (12)$$

$$\hat{q}_{ij,t,s}^+ + \hat{q}_{ij,t,s}^- = \sum_{l=1}^{\Lambda} \hat{\phi}_{ij,l,t,s}^{\hat{Q}} \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (13)$$

$$0 \leq \hat{\phi}_{ij,l,t,s}^{\hat{Q}} \leq \bar{\Phi}_{ij}^{\hat{S}} \quad \forall ij \in \Omega^B, l \in \{1, \dots, \Lambda\}, t \in \Omega^T, s \in \Omega^S \quad (14)$$

$$\hat{p}_{ij,t,s} = \sum_{a \in \Omega^C} p_{ij,a,t,s} \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (15)$$

$$\hat{q}_{ij,t,s} = \sum_{a \in \Omega^C} q_{ij,a,t,s} \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (16)$$

$$\hat{t}_{ij,t,s}^{SQ} = \sum_{a \in \Omega^C} t_{ij,a,t,s}^{SQ} \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (17)$$

$$0 \leq t_{ij,a,t,s}^{SQ} \leq \bar{I}_a^2 \chi_{ij,a,t}^C \quad \forall ij \in \Omega^B, a \in \Omega^C, t \in \Omega^T, s \in \Omega^S \quad (18)$$

$$\underline{V}^2 \leq v_{i,t,s}^{SQ} \leq \bar{V}^2 \quad \forall i \in \Omega^N, t \in \Omega^T, s \in \Omega^S \quad (19)$$

$$0 \leq p_{i,t,s}^{SS} \leq \bar{S}_i^{SS} \quad \forall i \in \Omega^{SS}, t \in \Omega^T, s \in \Omega^S \quad (20)$$

$$|q_{i,t,s}^{SS}| \leq \bar{S}_i^{SS} \quad \forall i \in \Omega^{SS}, t \in \Omega^T, s \in \Omega^S \quad (21)$$

$$|q_{i,t,s}^{SS}| \leq \sqrt{2} \bar{S}_i^{SS} - p_{i,t,s}^{SS} \quad \forall i \in \Omega^{SS}, t \in \Omega^T, s \in \Omega^S \quad (22)$$

Constraints (4)–(17) represent the application of Kirchhoff's laws to the distribution system operation [14]. Note that, (4) and (5) refer to the active and reactive power balances that

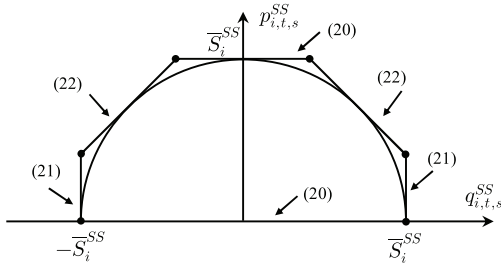


Fig. 3. Linearization of the capacity curve of the substations.

consider the polynomial ZIP load model to define the voltage-dependent behavior of the loads. The loads are modeled using constant impedance (Z), constant current (I), and constant power (P) components, according with the corresponding participation of each component, which is determined by Γ and Ψ [24]. Equation (6) calculates $v_{i,s} = \sqrt{v_{i,t,s}^{SQ}}$, used in the ZIP model, employing a Taylor series expansion at $(\bar{V} + \underline{V})/2$. Constraints (7) and (8) represent the application of Kirchhoff's voltage law to the system.

The original formulation for the DSP is an MINLP model and, therefore, to obtain an MILP model, the following considerations are made: (i) $v_{i,t,s}^2$ and $l_{ij,a,t,s}^2$ are linearized through variable replacements, i.e., $v_{i,t,s}^2$ by $v_{i,t,s}^{SQ}$, and $l_{ij,a,t,s}^2$ by $l_{ij,a,t,s}^{SQ}$. Note that $v_{i,t,s}^{SQ}$ and $l_{ij,a,t,s}^{SQ}$ are linear terms; (ii) $\hat{p}_{ij,t,s}^2$ and $\hat{q}_{ij,t,s}^2$ of the nonlinear equation $v_{j,t,s}^2 \hat{l}_{ij,t,s}^{SQ} = \hat{p}_{ij,t,s}^2 + \hat{q}_{ij,t,s}^2$, are linearized in (8)–(14) using a piecewise linearization in which the length of each block is $\bar{\Phi}_{ij} = \bar{V} \max_{a \in \Omega^C} \bar{I}_a / \Delta$, and the slope of each block is $\mathcal{M}_{ij,1}^{\hat{S}} = (5/6)\bar{\Phi}_{ij}$ and $\mathcal{M}_{ij,l}^{\hat{S}} = (2l - 1)\bar{\Phi}_{ij}$ for $l > 1$ [25].

Constraints (15)–(17) calculate, respectively, the active and reactive power flows and squared value of the current on branches according to the values on each conductor. Note that, the formulation requires that only one type of conductor is selected for each branch.

The currents on the branches and the voltages at the nodes are limited by (18) and (19), respectively. Constraints (20)–(22) are the capacity of the substations, which is illustrated in Fig. 3.

C. Constraints for Investment in Conductors and VRs

The set of constraints (23)–(26) involves the investment in the replacement of conductors.

$$\sum_{a \in \Omega^C} \sum_{t \in \Omega^T} \chi_{ij,a,t}^C \leq 1 \quad \forall ij \in \Omega^B \quad (23)$$

$$\hat{x}_{ij,a,t}^C = \sum_{h=1}^t \chi_{ij,a,h}^C \quad \forall ij \in \Omega^B, a \in \Omega^C, t \in \Omega^T \quad (24)$$

$$\check{x}_{ij,t}^C = \sum_{a \in \Omega^C} \hat{x}_{ij,a,t}^C \quad \forall ij \in \Omega^B, t \in \Omega^T \quad (25)$$

$$x_{ij,a,t}^C = \bar{x}_{ij,a}^C (1 - \check{x}_{ij,t}^C) + \hat{x}_{ij,a,t}^C \quad \forall ij \in \Omega^B, a \in \Omega^C, t \in \Omega^T \quad (26)$$

Constraint (23) guarantees the selection of a single type of conductor for a branch and that a conductor can only be replaced once in the planning horizon. Constraint (24) calculates the variable $\hat{x}_{ij,a,t}^C$, which indicates the type of the new conductor operating in each branch and stage. Constraint (25) calculates $\check{x}_{ij,t}^C$, showing in which stages the new conductor is operating. Finally, constraint (26) indicates the type of conductor, a , that is present on branch ij , $x_{ij,a,t}^C$, according to the new conductor type $\hat{x}_{ij,a,t}^C$ and the existing conductor type $\bar{x}_{ij,a}^C$.

Constraints (27)–(32) are related to the installation of VRs.

$$|\xi_{ij,t,s}^{VR}| \leq \mathcal{R}^{VR} (\mathcal{R}^{VR} + 2) v_{i,t,s}^{SQ} \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (27)$$

$$|\xi_{ij,t,s}^{VR}| \leq \sum_{a \in \Omega^C} x_{ij,a,t}^{VR} \mathcal{R}^{VR} (\mathcal{R}^{VR} + 2) \bar{V}^2 \quad \forall ij \in \Omega^B, t \in \Omega^T, s \in \Omega^S \quad (28)$$

$$\sum_{a \in \Omega^C} \sum_{t \in \Omega^T} \chi_{ij,a,t}^{VR} \leq 1 \quad \forall ij \in \Omega^B \quad (29)$$

$$x_{ij,a,t}^{VR} = \sum_{h=1}^t \chi_{ij,a,h}^{VR} \quad \forall ij \in \Omega^B, a \in \Omega^C, t \in \Omega^T \quad (30)$$

$$x_{ij,a,t}^{VR} \leq x_{ij,a,t}^C \quad \forall ij \in \Omega^B, a \in \Omega^C, t \in \Omega^T \quad (31)$$

$$\sum_{ij \in \Omega^B} \sum_{a \in \Omega^C} x_{ij,a,t}^{VR} \leq \bar{N}_t^{VR} \quad \forall t \in \Omega^T \quad (32)$$

Constraints (27) and (28) calculate $\xi_{ij,t,s}^{VR}$ for the branches, depending on the binary investment variable $x_{ij,a}^{VR}$. Constraint (29) requires that for a branch, a VR can only be installed at one stage, while (30) indicates the type of VR, a , that is present on branch ij , $x_{ij,a,t}^{VR}$, according to the investment variable $\chi_{ij,a,t}^{VR}$. Constraint (31) ensures that a VR type a can be installed in a branch only if it has a current capacity equal to the capacity of the conductor existing in that branch (note that the capacity of a conductor of type a must match the capacity of a VR of type a). Constraint (32) limits the total number of VRs that can be present in the distribution system at each stage.

D. Constraints for Investment in CBs

Constraints (33)–(45) are associated with the operation and investments in CBs.

$$\hat{q}_{i,t,s}^{CB} = \sum_{c=1}^{\bar{N}_i^{CB}} q_{i,c,t,s}^{CB} \quad \forall i \in \Omega^N, t \in \Omega^T, s \in \Omega^S \quad (33)$$

$$-\bar{V}^2 B^{CB} (1 - x_{i,c,t}^{CB}) \leq q_{i,c,t,s}^{CB} - v_{i,t,s}^{SQ} B^{CB} \leq -\underline{V}^2 B^{CB} (1 - x_{i,c,t}^{CB}) \quad (34)$$

$$\underline{V}^2 B^{CB} x_{i,c,t}^{CB} \leq q_{i,c,t,s}^{CB} \leq \bar{V}^2 B^{CB} x_{i,c,t}^{CB} \quad \forall i \in \Omega^N, c \in \{1, \dots, \bar{N}_i^{CB}\}, t \in \Omega^T, s \in \Omega^S \quad (35)$$

$$\sum_{c=1}^{\bar{N}_i^{CB}} x_{i,c,t}^{CB} = n_{i,t}^{CB} \quad \forall i \in \Omega^N, t \in \Omega^T \quad (36)$$

$$0 \leq n_{i,t}^{CB} \leq \bar{N}_i^{CB} y_{i,t}^{CB} \quad \forall i \in \Omega^N, t \in \Omega^T \quad (37)$$

$$\sum_{i \in \Omega^N} y_{i,t}^{CB} \leq \bar{N}_t^{CB} \quad \forall t \in \Omega^T \quad (38)$$

$$y_{i,t}^{CB} \geq y_{i,t-1}^{CB} \quad \forall i \in \Omega^N, t \in \Omega^T | t > 1 \quad (39)$$

$$x_{i,c,t}^{CB} \geq x_{i,c,t-1}^{CB} \quad \forall i \in \Omega^N, c \in \{1, \dots, \bar{N}_i^{CB}\}, t \in \Omega^T | t > 1 \quad (40)$$

$$x_{i,c-1,t}^{CB} \geq x_{i,c,t}^{CB} \quad \forall i \in \Omega^N, c \in \{2, \dots, \bar{N}_i^{CB}\}, t \in \Omega^T \quad (41)$$

$$\psi_{i,1}^{CB} = n_{i,1}^{CB} \quad \forall i \in \Omega^N \quad (42)$$

$$\psi_{i,t}^{CB} = n_{i,t}^{CB} - n_{i,t-1}^{CB} \quad \forall i \in \Omega^N, t \in \Omega^T | t > 1 \quad (43)$$

$$\chi_{i,1}^{CB} = y_{i,1}^{CB} \quad \forall i \in \Omega^N \quad (44)$$

$$\chi_{i,t}^{CB} = y_{i,t}^{CB} - y_{i,t-1}^{CB} \quad \forall i \in \Omega^N, t \in \Omega^T | t > 1 \quad (45)$$

Constraint (33) computes the total reactive power injected by a CB as the sum of the injections of each module, while (34) and (35) calculate the reactive power injection of each CB's module. Note that, when $x_{i,c,t}^{CB} = 1$, then $q_{i,c,t,s}^{CB} = B^{CB} y_{i,t,s}^{SQ}$ in (34), and when $x_{i,c,t}^{CB} = 0$, then $q_{i,c,t,s}^{CB} = 0$ in (35). Constraint (36) calculates the number of CB's modules installed at each node, constraint (37) limits the number of CB modules that can be installed at each node, and (38) restricts the number of CBs that can be present in the system at each stage. Constraints (39)–(41) prevent CBs to be removed from the system at stage t if they were installed at stage $t - 1$. Constraints (42) and (43) calculate the number of new CB modules installed in the system at each stage, while (44) and (45) calculate the number of new CBs installed at each stage.

E. Constraints for Investment in Dispatchable DG Units

Constraints (46)–(55) are related to the operation and investment in dispatchable DG units.

$$0 \leq p_{i,g,t,s}^G \leq \bar{S}_g^G x_{i,g,t}^G \quad \forall i \in \Omega^N, g \in \Omega^G, t \in \Omega^T, s \in \Omega^S \quad (46)$$

$$|q_{i,g,t,s}^G| \leq \bar{S}_g^G x_{i,g,t}^G \quad \forall i \in \Omega^N, g \in \Omega^G, t \in \Omega^T, s \in \Omega^S \quad (47)$$

$$|q_{i,g,t,s}^G| \leq \sqrt{2} \bar{S}_g^G x_{i,g,t}^G - p_{i,g,t,s}^G \quad \forall i \in \Omega^N, g \in \Omega^G, t \in \Omega^T, s \in \Omega^S \quad (48)$$

$$-p_{i,g,t,s}^G \tan\left(\cos^{-1}\left(\frac{PF_g^G}{\bar{PF}_g^G}\right)\right) \leq q_{i,g,t,s}^G \leq p_{i,g,t,s}^G \tan\left(\cos^{-1}\left(\frac{PF_g^G}{\bar{PF}_g^G}\right)\right) \quad \forall i \in \Omega^N, g \in \Omega^G, t \in \Omega^T, s \in \Omega^S \quad (49)$$

$$\sum_{g \in \Omega^G} x_{i,g,t}^G \leq 1 \quad \forall i \in \Omega^N, t \in \Omega^T \quad (50)$$

$$0 \leq x_{i,g,t}^G \leq \bar{x}_{i,g}^G \quad \forall i \in \Omega^N, g \in \Omega^G, t \in \Omega^T \quad (51)$$

$$\sum_{i \in \Omega^N} \sum_{g \in \Omega^G} x_{i,g,t}^G \leq \bar{N}_t^G \quad \forall t \in \Omega^T \quad (52)$$

$$x_{i,g,t}^G \geq x_{i,g,t-1}^G \quad \forall i \in \Omega^N, g \in \Omega^G, t \in \Omega^T | t > 1 \quad (53)$$

$$\chi_{i,g,1}^G = x_{i,g,1}^G \quad \forall i \in \Omega^N, g \in \Omega^G \quad (54)$$

$$\chi_{i,g,t}^G = x_{i,g,t}^G - x_{i,g,t-1}^G \quad \forall i \in \Omega^N, g \in \Omega^G, t \in \Omega^T | t > 1 \quad (55)$$

Constraints (46)–(49) represent the linearization of the capability curve of a synchronous machine, in which (49) shows the power factor limits of the generator. Constraint (50) ensures that, at maximum, a single type of dispatchable DG unit is installed at each node, (51) requires that dispatchable DG units can only be installed at candidate nodes, while (52) restricts the number of dispatchable DG units that can be present in the system at each stage. Constraint (53) ensures that dispatchable DG units are not removed from the system from one stage to another, and (54) and (55) calculate the number of new dispatchable DG units installed at each node at each stage.

F. Constraints for Investment in Nondispatchable RES Units

Constraints (56)–(66) are related to the operation and investment in nondispatchable RES units.

$$0 \leq p_{i,r,t,s}^{RS} \leq \bar{S}_r^{RS} \mathcal{F}_{r,s}^{RS} n_{i,r,t}^{RS} \quad (56)$$

$$|q_{i,r,t,s}^{RS}| \leq \bar{S}_r^{RS} \mathcal{F}_{r,s}^{RS} n_{i,r,t}^{RS} \quad (57)$$

$$|q_{i,r,t,s}^{RS}| \leq \sqrt{2} \bar{S}_r^{RS} \mathcal{F}_{r,s}^{RS} n_{i,r,t}^{RS} - p_{i,r,t,s}^{RS} \quad (58)$$

$$-p_{i,r,t,s}^{RS} \tan\left(\cos^{-1}\left(\frac{PF_r^{RS}}{\bar{PF}_r^{RS}}\right)\right) \leq q_{i,r,t,s}^{RS} \leq p_{i,r,t,s}^{RS} \tan\left(\cos^{-1}\left(\frac{PF_r^{RS}}{\bar{PF}_r^{RS}}\right)\right) \quad (59)$$

$$\forall i \in \Omega^N, r \in \Omega^{RS}, t \in \Omega^T, s \in \Omega^S \quad (59)$$

$$0 \leq n_{i,r,t}^{RS} \leq \bar{N}_{i,r}^{RS} y_{i,t}^{RS} \quad \forall i \in \Omega^N, r \in \Omega^{RS}, t \in \Omega^T \quad (60)$$

$$\sum_{i \in \Omega^N} n_{i,r,t}^{RS} \leq \bar{N}_{r,t}^{RS} \quad \forall r \in \Omega^{RS}, t \in \Omega^T \quad (61)$$

$$n_{i,r,t}^{RS} \geq n_{i,r,t-1}^{RS} \quad \forall i \in \Omega^N, r \in \Omega^{RS}, t \in \Omega^T | t > 1 \quad (62)$$

$$\psi_{i,r,1}^{RS} = n_{i,r,1}^{RS} \quad \forall i \in \Omega^N, r \in \Omega^{RS} \quad (63)$$

$$\psi_{i,r,t}^{RS} = n_{i,r,t}^{RS} - n_{i,r,t-1}^{RS} \quad \forall i \in \Omega^N, t \in \Omega^T | t > 1 \quad (64)$$

$$\chi_{i,r,1}^{RS} = y_{i,r,1}^{RS} \quad \forall i \in \Omega^N, r \in \Omega^{RS} \quad (65)$$

$$\chi_{i,r,t}^{RS} = y_{i,r,t}^{RS} - y_{i,r,t-1}^{RS} \quad \forall i \in \Omega^N, r \in \Omega^{RS}, t \in \Omega^T | t > 1 \quad (66)$$

Constraints (56)–(59) represent the capacity of the power converter of the nondispatchable RES units of type r , in which (59) limits its power factor. Constraint (60) limits the number of the nondispatchable RES units of type r that can be installed at each node, (61) limits the number of the nondispatchable RES units that can be present in the system at each stage, (62) ensures that the nondispatchable RES units are not removed from the system from one stage to another, and (63) and (64) calculate the number of new nondispatchable RES units of type r installed in each stage. Finally, (65) and (66) calculate the number of new nondispatchable RES units of type r were installed at each stage.

G. Constraints for Investment in ES Units

Constraints (67)–(76) are related to the ES units.

$$0 \leq p_{i,e,t,s}^{ES-} \leq n_{i,e,t}^{ES} \bar{P}_e^{ES} \quad \forall i \in \Omega^N, e \in \Omega^{ES}, t \in \Omega^T, s \in \Omega^S \quad (67)$$

$$0 \leq p_{i,e,t,s}^{ES+} \leq n_{i,e,t}^{ES} \bar{P}_e^{ES} \quad \forall i \in \Omega^N, e \in \Omega^{ES}, t \in \Omega^T, s \in \Omega^S \quad (68)$$

$$\sum_{s \in \Omega^S | \mathcal{G}_s = w} \Delta_s^T \left[\Xi_e^{ES-} p_{i,e,t,s}^{ES-} - \left(\frac{1}{\Xi_e^{ES+}} \right) p_{i,e,t,s}^{ES+} \right] = 0$$

$$\forall i \in \Omega^N, e \in \Omega^{ES}, w \in \Omega^{GS}, t \in \Omega^T \quad (69)$$

$$0 \leq n_{i,e,t}^{ES} \leq \bar{N}_{i,e}^{ES} y_{i,t}^{ES} \quad \forall i \in \Omega^N, e \in \Omega^{ES}, t \in \Omega^T \quad (70)$$

$$\sum_{i \in \Omega^N} n_{i,e,t}^{ES} \leq \bar{N}_{e,t}^{ES} \quad \forall e \in \Omega^{ES}, t \in \Omega^T \quad (71)$$

$$n_{i,e,t}^{ES} \geq n_{i,e,t-1}^{ES} \quad \forall i \in \Omega^N, e \in \Omega^{ES}, t \in \Omega^T | t > 1 \quad (72)$$

$$\psi_{i,e,1}^{ES} = n_{i,e,1}^{ES} \quad \forall i \in \Omega^N, e \in \Omega^{ES} \quad (73)$$

$$\psi_{i,e,t}^{ES} = n_{i,e,t}^{ES} - n_{i,e,t-1}^{ES} \quad \forall i \in \Omega^N, e \in \Omega^{ES}, t \in \Omega^T | t > 1 \quad (74)$$

$$\chi_{i,e,t}^{ES} = y_{i,e,1}^{ES} \quad \forall i \in \Omega^N, e \in \Omega^{ES} \quad (75)$$

$$\chi_{i,e,t}^{ES} = y_{i,e,t}^{ES} - y_{i,e,t-1}^{ES} \quad \forall i \in \Omega^N, e \in \Omega^{ES}, t \in \Omega^T | t > 1 \quad (76)$$

Constraints (67) and (68) represent the capacity of the converter for charging and discharging an ES unit of type e , constraint (69) defines the transition function of ES units to be accomplished at each group of scenarios, i.e., the energy can only be exchanged (stored or injected) among the scenarios of the same group (each group representing a season of the year) in each stage [26]. Constraint (70) restricts the number of ES units of type e that can be installed at each node, (71) limits the number of ES units that can be present in the system at each stage, constraint (72) ensures that ES units are not removed from the system from one stage to another, and (73) and (74) calculate the number of new ES units of type e installed. Finally, (75) and (76) calculate the number of new nodes in which ES units of type e were installed at each stage.

H. Constraints for Investment in EVCSs

The set of constraints (77)–(85) comprises the investment in the EVCSs.

$$0 \leq p_{i,v,t,s}^{EV} \leq \bar{P}_v^{EV} n_{i,v,t}^{EV} \quad \forall i \in \Omega^N, v \in \Omega^{EV}, t \in \Omega^T, s \in \Omega^S \quad (77)$$

$$\sum_{i \in \Omega^N | \mathcal{Z}_i = z} p_{i,v,t,s}^{EV} = \mathcal{P}_{z,v,t,s}^{EV} \quad \forall z \in \Omega^Z, v \in \Omega^{EV}, t \in \Omega^T, s \in \Omega^S \quad (78)$$

$$\sum_{v \in \Omega^{EV}} n_{i,v,t}^{EV} \leq \bar{N}_i^{EV} y_{i,t}^{EV} \quad \forall i \in \Omega^N, t \in \Omega^T \quad (79)$$

$$n_{i,v,t}^{EV} \geq n_{i,v,t-1}^{EV} \quad \forall i \in \Omega^N, v \in \Omega^{EV}, t \in \Omega^T | t > 1 \quad (80)$$

$$y_{i,t}^{EV} \geq y_{i,t-1}^{EV} \quad \forall i \in \Omega^N, t \in \Omega^T | t > 1 \quad (81)$$

$$\psi_{i,v,1}^{EV} = n_{i,v,1}^{EV} \quad \forall i \in \Omega^N, v \in \Omega^{EV} \quad (82)$$

$$\psi_{i,v,t}^{EV} = n_{i,v,t}^{EV} - n_{i,v,t-1}^{EV} \quad \forall i \in \Omega^N, v \in \Omega^{EV}, t \in \Omega^T | t > 1 \quad (83)$$

$$\chi_{i,1}^{EV} = y_{i,1}^{EV} \quad \forall i \in \Omega^N \quad (84)$$

$$\chi_{i,t}^{EV} = y_{i,t}^{EV} - y_{i,t-1}^{EV} \quad \forall i \in \Omega^N, t \in \Omega^T | t > 1 \quad (85)$$

Constraints (77) represents the capacity of the EVs' charger of type v , e.g., fast and slow chargers, constraints (78) defines that each zone must have a required installed capacity of EVs' charger of type v , constraint (79) requires the installation of the

infrastructure for EVCSs, constraints (80)–(81) ensures that EVs' chargers and EVCSs, respectively, are not removed from the system from one stage to another, while (82)–(85) calculate the number of new chargers of type v and EVCSs installed in each stage.

It is considered that the infrastructure for EVCSs can be installed only at candidate nodes of the distribution system. These candidate nodes are determined according to their proximity to the vehicular traffic network.

I. DERs' Penetration Requirement and CO₂ Emissions Limit

Constraints (86) and (87) model, respectively, the minimum penetration of DERs and the CO₂ emissions limit.

$$\sum_{i \in \Omega^N} \sum_{s \in \Omega^S} \Delta_s^T \left(\sum_{g \in \Omega^G} p_{i,g,t,s}^G + \sum_{r \in \Omega^{RS}} p_{i,r,t,s}^{RS} + \sum_{e \in \Omega^{ES}} p_{i,e,t,s}^{ES+} \right) \geq \mathcal{U}_t \left[\sum_{i \in \Omega^N} \sum_{s \in \Omega^S} \Delta_s^T \left(p_{i,t,s}^D + \sum_{e \in \Omega^{ES}} p_{i,e,t,s}^{ES-} + \sum_{v \in \Omega^{EV}} p_{i,v,t,s}^{EV} \right) + \sum_{ij \in \Omega^B} \sum_{s \in \Omega^S} \Delta_s^T R_{aLij}^{SQ} \right] \quad \forall t \in \Omega^T \quad (86)$$

$$\sum_{i \in \Omega^N} \sum_{t \in \Omega^T} \sum_{s \in \Omega^S} (\Upsilon_t^F - \Upsilon_t^I) \Delta_s^T \times \left(\mathcal{E}_i^{SS} p_{i,t,s}^{SS} + \sum_{g \in \Omega^G} \mathcal{E}_g^G p_{i,g,t,s}^G + \sum_{r \in \Omega^{RS}} \mathcal{E}_r^{RS} p_{i,r,t,s}^{RS} \right) \leq \Theta \quad (87)$$

Constraint (86) defines that a minimum ratio $\mathcal{U}_t$ of the energy supplied must be provided by DERs at stage t . Constraint (87) limits the total CO₂ emissions from the system in all stages.

IV. TESTS AND NUMERICAL RESULTS

The proposed model was written in AMPL [34] and solved with CPLEX v20.1.0.0 [35], with default settings, on a computer with a 3.20 GHz Intel Core i7–8700 processor and 32 GB of RAM. Two distribution systems were used to test the proposed model: a 69-node test system and a 134-node real system.

A. 69-Node Distribution System

The 69-node distribution system from [27] was adapted and used to test the proposed model. The system topology is depicted in Fig. 4, which includes 1 substation node, 66 load nodes, 2 transfer nodes, and 68 branches. Type I conductors are present in all branches of the network. The nominal voltage of the system is 12.66 kV. The lower and upper voltage magnitude limits are 0.95 p.u. and 1.05 p.u., respectively. The planning horizon is 6 years, divided into 3 stages of 2 years each. It is considered that 1,000 EVs are adopted in the initial year of the planning horizon. The demand of the system is estimated based on the annual average growth of 3% and an annual increase in the average penetration of EVs of 10%.

TABLE II
INVESTMENT PLANS

	Case I	Case II	Case III	Case IV
Conductors (branch ij , type of conductor, stage)	01-02, III, 1; 02-03, III, 1; 03-04, III, 1; 04-05, III, 1; 08-09, III, 1; 05-06, III, 2; 06-07, III, 2; 07-08, III, 2	01-02, III, 1; 02-03, III, 1; 03-04, III, 1; 08-09, III, 1; 04-05, III, 2; 05-06, III, 2; 07-08, III, 2	01-02, III, 1; 02-03, III, 1; 08-09, III, 1; 03-04, III, 2; 04-05, III, 2; 05-06, III, 2; 07-08, III, 2	01-02, III, 1; 02-03, III, 1; 03-04, III, 1; 08-09, III, 1; 04-05, III, 2; 05-06, III, 2; 07-08, III, 3
VRs (branch ij , type of VR, stage)	16-17, I, 3	06-07, I, 1; 16-17, I, 1; 03-28, I, 1; 03-36, I, 1; 09-53, I, 1; 59-60, I, 1	06-07, I, 1; 16-17, I, 1; 03-28, I, 1; 03-36, I, 1; 09-53, I, 1; 59-60, I, 1	06-07, I, 1; 16-17, I, 1; 03-28, I, 1; 03-36, I, 1; 09-53, I, 1; 59-60, I, 1
CBs (node i , number of units, stage)	19, 1, 1; 22, 1, 1; 60, 1, 1	19, 1, 1; 25, 1, 1; 60, 1, 1	19, 1, 1; 25, 1, 1; 60, 1, 1; 60, 1, 3	19, 1, 1; 25, 1, 1; 60, 1, 1
Biomass DG units (node i , type of DG, stage)	49, II, 1; 50, II, 1; 61, II, 1; 64, II, 1	49, II, 1; 50, II, 1; 61, II, 1; 64, II, 1	49, II, 1; 50, II, 1; 61, II, 1; 64, II, 1	49, II, 1; 50, II, 1; 61, II, 1; 64, II, 1
PV units (node i , number of units, stage)	21, 2, 2; 12, 3, 3; 16, 3, 3; 17, 3, 3; 18, 3, 3; 21, 1, 3	21, 3, 2; 12, 3, 3; 16, 3, 3; 17, 3, 3; 18, 3, 3	21, 3, 2; 12, 3, 3; 16, 3, 3; 17, 3, 3; 18, 3, 3	21, 3, 1; 12, 3, 2; 16, 3, 2; 17, 3, 2; 18, 3, 2; 07, 3, 3; 08, 3, 3; 11, 3, 3; 51, 3, 3; 59, 1, 3
WT units (node i , number of units, stage)	08, 3, 1; 51, 3, 1	08, 3, 1; 51, 3, 1	08, 3, 1; 51, 3, 1	08, 3, 1; 51, 3, 1
ES units (node i , number of units, stage)	07, 3, 1; 21, 3, 1; 07, 1, 2; 08, 4, 2; 11, 1, 2; 21, 1, 2; 51, 3, 2; 11, 2, 3; 48, 2, 3; 51, 1, 3	21, 3, 1; 07, 4, 2; 08, 3, 2; 21, 1, 2; 08, 1, 3; 48, 1, 3; 51, 4, 3	21, 3, 1; 07, 4, 2; 08, 3, 2; 21, 1, 2; 08, 1, 3; 11, 1, 3; 51, 4, 3	21, 3, 1; 07, 2, 2; 21, 1, 2; 07, 2, 3; 08, 4, 3; 51, 1, 3
Slow EV charger (node i , number of units, stage)	47, 6, 1; 62, 10, 1; 05, 5, 2; 47, 10, 2; 05, 4, 3; 51, 11, 3	47, 6, 1; 62, 10, 1; 05, 5, 2; 47, 10, 2; 05, 5, 3; 51, 10, 3	17, 4, 1; 30, 4, 1; 39, 3, 1; 47, 4, 1; 62, 3, 1; 17, 3, 2; 30, 3, 2; 39, 3, 2; 47, 4, 2; 62, 3, 2; 17, 3, 3; 30, 3, 3; 39, 2, 3; 47, 3, 3; 62, 2, 3	17, 4, 1; 30, 4, 1; 39, 3, 1; 47, 4, 1; 62, 3, 1; 17, 3, 2; 30, 3, 2; 39, 3, 2; 47, 4, 2; 62, 3, 2; 17, 3, 3; 30, 3, 3; 39, 2, 3; 47, 3, 3; 62, 2, 3
Fast EV charger (node i , number of units, stage)	47, 2, 1; 62, 7, 1; 05, 3, 2; 47, 2, 2; 62, 3, 2; 05, 4, 3; 51, 4, 3	47, 2, 1; 62, 7, 1; 05, 3, 2; 47, 2, 2; 62, 3, 2; 05, 5, 3; 51, 3, 3	17, 2, 1; 30, 2, 1; 39, 2, 1; 47, 3, 1; 62, 2, 1; 17, 2, 2; 30, 2, 2; 39, 2, 2; 47, 2, 2; 62, 1, 2; 17, 2, 3; 30, 1, 3; 39, 1, 3; 47, 2, 3; 62, 1, 3	17, 2, 1; 30, 2, 1; 39, 2, 1; 47, 3, 1; 62, 2, 1; 17, 2, 2; 30, 2, 2; 39, 2, 2; 47, 2, 2; 62, 1, 2; 17, 2, 3; 30, 1, 3; 39, 1, 3; 47, 2, 3; 62, 1, 3

*The highlighted areas show the main differences in investments made in the network for each one of the case studies considered in this work with respect to the previous case.

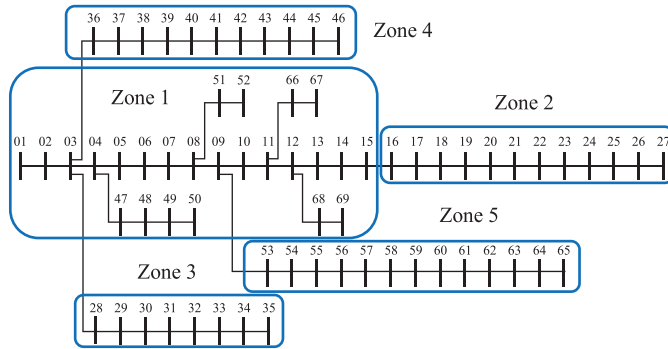


Fig. 4. 69-node distribution system divided by zones.

The number of centroids is $k = 6$, thereby obtaining 48 operation scenarios. The annual interest rate is 10%. The number of blocks in the piecewise linearization is $\Lambda = 10$. The criterion for charging EVs is $\mu = 30\%$. The DERs candidate to be installed in the system include dispatchable biomass DG units, nondispatchable PV and WT units, and battery ES units. Both fast and slow chargers are considered for EVCSs. Complete data is available in [28].

Four case studies have been analyzed in order to assess the relevance of the proposed approach. In Case I, the DSP is developed using a constant power load model, thereby disregarding the voltage-dependent load model. In addition, the division of the system's concession area by zones as well as the limit on CO₂ emissions are also ignored. Case II is similar to Case I, the only difference is that in Case II a voltage-dependent load model is used. In Case III, the DSP is developed using a voltage-dependent load model, and considering the division of the system's concession area by zones, but without considering a limit on CO₂ emissions. Case IV is

similar to Case III, the difference is that Case IV considers a limit on CO₂ emissions. The technical details of the investment alternatives considered in the four case studies are available in [28].

The computational times to solve Cases I–IV are 16.70 h, 26.54 h, 6.32 h, and 10.93 h, respectively. The system concession area is divided into five zones, as shown in Fig. 4, for the implementation of Cases III and IV, but, for real case studies, it is suggested to use the division of the city map according to census tracts, as stated in [16]. Moreover, to obtain the total number of EVs in each zone, it is necessary to carry out an economic feasibility study or an EVs purchase intention survey to know the interest of people in adopting this new technology [16]. Table II shows the investment plans for the four cases, while the investment costs and total CO₂ emissions associated with each one of the planning strategies are reported in Table III.

The importance of modeling the load as voltage-dependent is assessed in two analyses. In the first analysis, the effect of such modeling is examined on the investment cost of the DSP. For this purpose, the investment plans obtained for Cases I and II are compared. Table III shows that modeling the load as voltage-dependent results in an investment strategy that is 5.49% cheaper than the investment strategy achieved by modeling the load as constant power. In addition, Table III shows that the operation cost associated with the investment strategy of Case II is 7.66% cheaper than the operation cost associated with the investment strategy of Case I. Thus, the total cost of Case II is 7.36% cheaper than the total cost of Case I.

Table III also shows that the CO₂ emissions associated with Case II are 9.05% lower than the CO₂ emissions associated

TABLE III
OVERVIEW OF THE SOLUTIONS

Results		Case I	Case II	Case III	Case IV
Investment cost (10 ³ USD)	Conductors	15.15	12.24	11.74	11.32
	CBs	10.62	11.30	11.30	10.62
	VRs	1.44	37.91	37.91	37.91
	Biomass DG units	171.19	171.19	171.19	171.19
	PV units	137.69	148.65	148.65	426.17
	WT units	485.64	485.64	485.64	485.64
	ES units	404.71	280.74	280.74	199.88
	EVCS installation	88.91	88.91	159.99	159.99
	Fast charger	57.11	57.11	65.82	65.82
	Slow charger	62.28	62.28	66.99	66.99
Total investment cost (10 ³ USD)		1,434.74	1,355.97	1,439.97	1,635.53
Total operation cost (10 ³ USD)		8,925.54	8,241.79	8,258.63	8,102.04
Total cost (10 ³ USD)		10,360.28	9,597.76	9,698.60	9,737.57
CO ₂ emissions (10 ³ tonnes)		75.92	69.05	69.16	65.00

with Case I. On the other hand, Table II shows that the investment plan obtained for Case II gives priority to the installation of VRs, since, by modeling the load as voltage-dependent, it is possible to reduce the network voltage levels to reduce the load and thus decrease energy consumption. Moreover, Table II shows that, by modeling the load as voltage-dependent, the number of equipment needed for the network to operate correctly decreases.

In a second analysis, the impact of modeling the load as voltage-dependent is assessed on the operation costs of the DSP. For this purpose, the investment plan obtained for Case I is fixed, and the operation of the system is optimized considering the voltage-dependent load model. In this case, the operation cost is $8,427.14 \times 10^3$ USD, which is 2.25% higher than the operation cost of Case II, and the CO₂ emissions are 72.02×10^3 tonnes, which is 4.30% higher than the CO₂ emissions of Case II.

The relevance of planning EVCSs by zones can be analyzed by comparing Cases II and III. As can be observed in Table III, an increase in the investment cost associated with the deployment of EVCSs is attained for Case III, thereby leading to an increase of 1.05% in the total cost. Note that, as can be seen in Tables II and III, Case III provides more EVCSs coverage for charging EVs' batteries, with 67% more EVCSs available at the end of the planning horizon than Case II. It is also worth mentioning that the investment strategy obtained for Case III is better (cheaper and more sustainable) than the solution obtained for Case I. Case I represents the way in which this problem has been commonly approached [7]–[10].

Finally, the importance of incorporating CO₂ emissions constraints can be analyzed by comparing Case III with Case IV. As can be observed in Table II, when the CO₂ emissions constraint is considered, the solution process prioritizes the installation of renewable sources, such as PV units. Furthermore, Table III shows that considering CO₂ emissions constraints results in a plan that has a total cost of 0.40% more expensive but with 6.02% less CO₂ emissions. By reducing the CO₂ emissions limit by 15.38% in relation to Case IV, from 65.00×10^3 tonnes to 55.00×10^3 tonnes, the investment cost increases 49.78%, from $1,635.53 \times 10^3$ USD to $2,449.65 \times 10^3$ USD, while the operation cost decreases 1.30%,

TABLE IV
IMPACT OF THE NUMBER OF EVs ADOPTED

Total number of EVs	Total number of EVCSs	Number of chargers		Total planning cost (10 ³ USD)	Difference of the cost (%)
		Fast	Slow		
600	5	12	21	9,345.73	−4.02
700	5	17	24	9,409.41	−3.37
800	5	21	34	9,543.75	−1.99
900	5	27	44	9,670.23	−0.69
1,000	5	27	47	9,737.57	–
1,100	6	33	53	9,831.22	0.96
1,200	6	34	58	9,901.75	1.69
1,300	8	39	65	10,002.73	2.72
1,400	8	42	68	10,058.58	3.30

TABLE V
IMPACT OF THE NUMBER OF ZONES

Total number of zones	Total number of EVCSs	Number of chargers		Total planning cost (10 ³ USD)	Difference of the cost (%)
		Fast	Slow		
3	3	26	47	9,680.27	−0.59
4	4	27	47	9,694.93	−0.44
5	5	27	47	9,737.57	–
6	6	26	49	9,768.43	0.32
7	7	28	48	9,806.93	0.71

from $8,102.04 \times 10^3$ USD to $7,997.05 \times 10^3$ USD. This type of analysis can help the system operator better understand the relationship between the extra costs and CO₂ emissions reduction.

A sensitivity analysis with different values of EVs adopted has been carried out in order to calculate the variation in the total planning cost since it is difficult to accurately predict the adoption rate of EVs. This rate is influenced by different factors such as national subsidy policies for the purchase of EVs, the performance, and autonomy of batteries, the cost of purchasing and maintaining EVs, the availability of infrastructure for charging batteries, etc. This sensitivity analysis has been carried out considering the conditions established for Case IV. Table IV shows that the total number of EVs adopted has a direct impact on the number of EVCS, the number of fast and slow chargers, and the total cost. Specifically, as can be observed, a reduction of 40% in the number of EVs adopted results in a reduction of 4.02% in total cost, while an increase of 40% in the number of EVs adopted results in an increase of 3.30% in total cost. The computational times to obtain the results shown in Table IV are between 9.85 h and 13.70 h.

Another sensitivity analysis has been carried out increasing and decreasing the number of zones into which the distribution system's operating area is divided, in order to calculate the influence of the number of zones on the total planning cost. It should be noted that the proposed model does not optimize the number of zones into which the system is divided because that information is input data. This sensitivity analysis has been carried out considering a total of 1,000 EVs adopted at the first stage and the conditions established for Case IV. Table V shows that the division of the network coverage area into zones has a direct impact on the total number of EVCSs built. Moreover, as can be observed, the total number of fast and slow chargers remains almost constant for

different numbers of zones, but such chargers are more distributed across the system when larger numbers of EVCSs are built. This fact is positive because the availability of the battery charging infrastructure is one of the factors that have the biggest impact on promoting the adoption of EVs. In addition, planning the EVCS by zones allows knowing the number of fast and slow chargers that must be installed in each zone to supply the EV charging demand. For the results shown in Table V, the computational times to solve the problem with 3, 4, 5, 6, and 7 zones are 31.09 h, 12.96 h, 10.93 h, 11.13 h, and 12.81 h, respectively. These results show that, by considering larger numbers of zones in the system, the computational times to solve the problem decrease in relation to the computational times to solve the problem without considering zones (Case II), or with a small number of zones (when 3 zones are considered), because when this division is made, the problem becomes more restricted, and thus the feasible region of the solution is reduced.

B. 134-Node Distribution System

To validate the scalability of the proposed model, results for a 134-node distribution system are presented. This system consists of 1 substation node, 127 load nodes, 6 transfer nodes, and 133 branches. The nominal voltage of the system is 14.40 kV. The lower and upper voltage magnitude limits are 0.95 p.u. and 1.05 p.u., respectively. The planning horizon is 6 years, divided into 3 stages of 2 years each. It is considered that 1,500 EVs are adopted in the initial year of the planning horizon. The demand of the system is estimated based on the annual average growth of 3% and an annual increase in the average penetration of EVs of 10%. The annual interest rate is 10%. The number of blocks in the piecewise linearization is $\Lambda = 10$. The DERs candidate to be installed in the system include dispatchable biomass DG units, nondispatchable PV and WT units, and battery ES units. Both fast and slow chargers are considered for EVCSs. The network was divided into nine zones. Complete data is available in [27].

A sensitivity analysis with different values for the limit of CO₂ emissions has been carried out to calculate the variation in the total investment and operation costs. Such sensitivity analysis is carried out only for the conditions established in Case IV for the 69-node system, i.e., considering the voltage-dependent load model, the division of the system's concession area by zones, and considering a limit on CO₂ emissions.

Fig. 5 shows that when CO₂ emissions limits increase, the total investment cost decreases, and the total operation cost increases. This fact is explained by the fact that, for reducing CO₂ emissions, the penetration of DERs must increase, and the investment costs in DERs are high while their operation costs are low when compared to the energy price at substations.

With the available investment alternatives, the minimum value for CO₂ emissions is 65.00×10^3 tonnes with an investment cost of $16,051.94 \times 10^3$ USD and an operation cost of $13,310.02 \times 10^3$ USD. On the other hand, without considering the limit for CO₂ emissions, a solution with total emissions of 178.00×10^3 tonnes is obtained, an investment cost of $2,446.18 \times 10^3$ USD and an operation cost of $21,725.06 \times 10^3$

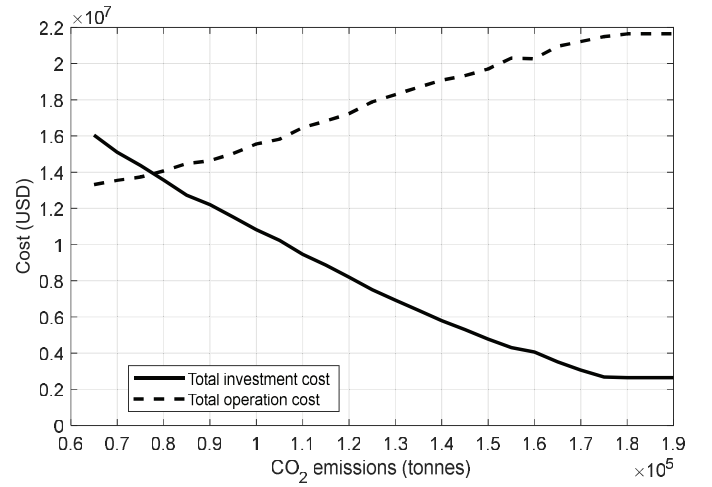


Fig. 5. Impact of CO₂ emissions on investment and operation costs.

USD. Therefore, the results indicate that for reducing the CO₂ emissions by 63.48%, from 178.00×10^3 tonnes to 65.00×10^3 tonnes, the total investment cost increased 556.20%, while the total operation cost was reduced by 38.73%. Moreover, the total cost increased 21.47%, from $24,171.24 \times 10^3$ USD to $29,361.96 \times 10^3$ USD.

These findings imply that as more money is invested in renewable DERs, the lower are the values of CO₂ emissions that can be obtained. As for the computational times, the solutions shown in Fig. 5 requires 47.18 h on average, which are suitable for a planning context.

V. CONCLUSION

A multistage mixed-integer linear programming model is proposed in this paper to address the problem of medium-term planning of active distribution systems and electric vehicle charging stations (EVCSs). The model considers several investment alternatives, such as the replacement of conductors, installation of capacitor banks, voltage regulators, dispatchable and renewable distributed generation, and energy storage units. The uncertainties, e.g., wind speed, solar irradiation, electricity demand, energy prices, and EVCSs load profiles, are characterized through a set of scenarios. A voltage-dependent load representation is used to investigate the advantages of this approach in the results of the planning strategy. Besides that, the proposed model considers the installation of EVCSs by zones to improve the coverage of the EVCSs, and constraints related to CO₂ emissions to guarantee a sustainable distribution system.

The results show that the use of a voltage-dependent load model significantly reduces energy consumption compared to the use of a constant load model. Such decrease can be seen in the cost of the energy supplied by the substation. There is also a decrease in CO₂ emissions, which ensures that more environmentally sustainable and affordable initiatives can be accomplished by the use of an appropriate load representation.

The planning of EVCSs considering zones leads to higher investment costs, but it guarantees a greater coverage of electric vehicles (EVs) charging, which is reflected in the results

with a larger number of chargers installed. Finally, when considering CO₂ emissions, a more expensive investment plan is obtained, but one that can help avoid penalties for exceeding the allowable emissions limits. The investment plan obtained helps to prevent the charging of EVs from becoming a problem of grid emissions.

Future works will address the subject of decentralized planning with multiple agents, the possibility of incorporating information on the vehicular traffic network explicitly in the optimization model, and the vehicle-to-grid operation of EVs.

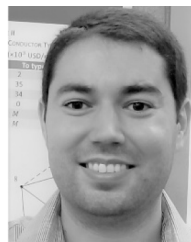
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